Machine Learning Mini-Project

Predicting Hawaii Coral Reef Ecological Regimes

Project Report | EDA, Reference-Model Replication, and K-Nearest Neighbors Extension

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1. Problem Statement

Coral reef ecological regimes reflect the combined influence of human pressures and environmental conditions. This project uses environmental and human-activity predictors to classify each Hawaii reef observation into one of four regime labels: 1, 2, 3, or 5. The work has two parts: exploratory data analysis and replication of reference classifiers, followed by an extension using K-Nearest Neighbors (KNN).

The primary comparison metric is Micro-F1, matching the project notebooks. In single-label multiclass classification, micro-averaged F1 equals overall accuracy; the report still labels the metric Micro-F1 to preserve the project's stated evaluation convention.

2. Dataset and Preprocessing

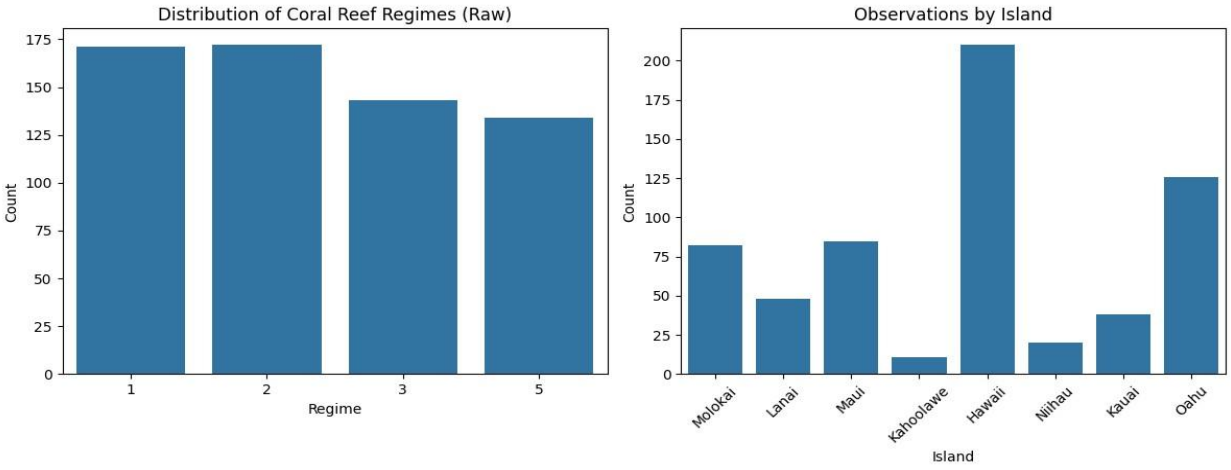
The raw dataset contains 620 spatial observations across Hawaii, including location/island information, reef/community measurements, human-impact variables, environmental variables, and the target regime. The raw EDA notebook identifies 39 columns in the raw table, including metadata and the target.

Predictors used for modelling: 25 numeric features consisting of 20 original predictors and five engineered features. The original predictors include effluent, sedimentation, development, habitat modification, invasive algae, several fishing-pressure measures, sea-surface temperature, chlorophyll, irradiance, wave conditions, complexity, and depth. The engineered predictors encode selected interactions/ratios among environmental factors.

The raw EDA reports missing values in Complexity (113 records) and Depth (12 records). For model training, the notebooks load the project's pre-imputed and pre-engineered train/validation/test files; the resulting training and test feature matrices contain no missing values.

Split: the provided training and validation portions are combined to form 496 training observations (80%); 124 observations (20%) are held out for final testing. The KNN notebook checks these sizes with assertions.

Exploratory view: regime and island distributions



The raw regime distribution is moderately balanced: regime 1 has 171 observations (27.58%), regime 2 has 172 (27.74%), regime 3 has 143 (23.06%), and regime 5 has 134 (21.61%). Micro-F1 is used consistently with the source project rather than changing the evaluation metric.

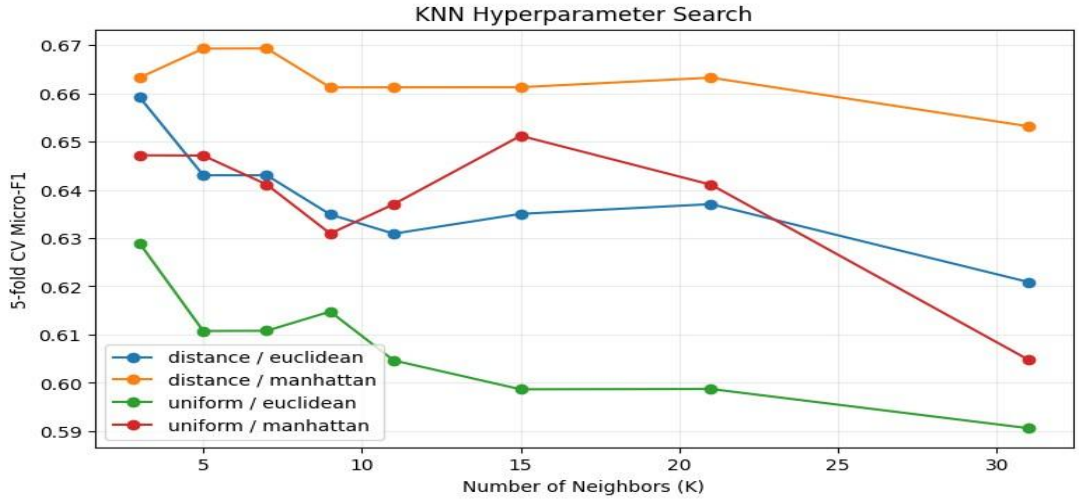
3. Methodology

Reference models. Part 1 fits a scaled Logistic Regression classifier and a Random Forest classifier. Logistic Regression is placed in a StandardScaler pipeline; the Random Forest uses scikit-learn's RandomForestClassifier with random_state=42. Both are evaluated with five-fold cross-validation using Micro-F1 and then scored on the held-out test set.

KNN extension. Part 2 implements a pipeline containing StandardScaler followed by KNeighborsClassifier. Scaling is important because KNN bases predictions on feature-space distances.. The best configuration is selected using 5-fold CV Micro-F1, not by choosing based on test performance.

Best KNN parameters: n_neighbors = 7, metric = Manhattan, weights = distance. The best cross-validation Micro-F1 is 0.6693.

KNN hyperparameter search



4. Results

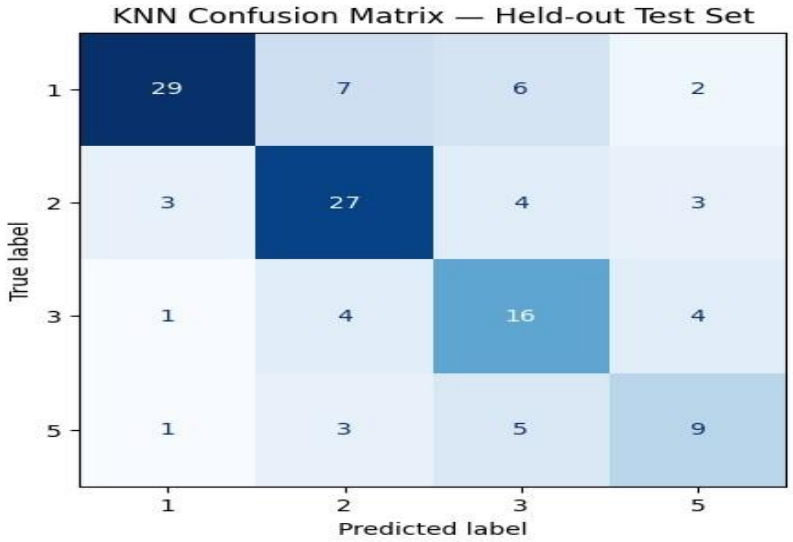
The replicated reference results and the KNN extension are summarized below. Scores are taken from the repository's saved model_comparison.csv and notebook outputs.

Model	5-fold CV Micro-F1	Test Micro-F1
Logistic Regression (replicated)	0.6189	0.5806
Random Forest (replicated)	0.6754	0.7258
KNN (extension)	0.6693	0.6532

Interpretation. KNN improves on Logistic Regression by about 7.26 percentage points in test Micro-F1 (0.6532 vs 0.5806), but it does not outperform Random Forest, which achieves 0.7258. KNN's CV score (0.6693) is close to Random Forest's CV score (0.6754), but the test result is lower. This difference suggests that strong cross-validation performance alone does not guarantee the best held-out performance.

The best KNN training Micro-F1 is 1.0000, while the test score is 0.6532. Because distance-weighted KNN can reproduce labels of training points when each point is among its own nearest neighbors, the perfect training score should not be interpreted as evidence of generalization; the test score is the more relevant estimate for unseen observations.

KNN confusion matrix on the held-out test set



5. Discussion

The KNN extension is a valid alternative model and follows the project's comparison setup: the same prepared features, the same train/test split, the same Micro-F1 metric, and training-only cross-validation for hyperparameter selection. Standardization and distance-based weighting are important components of the implementation.

The results indicate that the reef-regime classification problem benefits from more than simple proximity in standardized feature space. Random Forest performs best on the held-out set, consistent with the possibility that nonlinear interactions and threshold-like relationships among ecological variables matter. This is a plausible interpretation of the observed scores, not a causal conclusion established by the experiment.

6. Limitations and Reproducibility Notes

- The raw dataset has missing values in Complexity and Depth; modelling uses the preprocessed data supplied by thesource project rather than reproducing the full imputation process within the KNN notebook.
- The dataset contains 620 spatial observations from Hawaii. The report does not establish that the same scores generalizeto other reef systems or future observations.
- Training Micro-F1 of 1.0000 for distance-weighted KNN is not a useful stand-alone performance measure; test andcross-validation scores are more informative.

7. Conclusion

This project builds a reproducible comparison of Logistic Regression, Random Forest, and an added KNN classifier for predicting Hawaii coral reef regimes. The best KNN configuration uses seven neighbors, Manhattan distance, and distance weighting, achieving cross-validation Micro-F1 of 0.6693 and held-out test Micro-F1 of 0.6532. It improves on the replicated Logistic Regression baseline (0.5806 test Micro-F1), but remains below Random Forest (0.7258). KNN is therefore a useful extension, but Random Forest is the strongest of the three evaluated models on this test split. Future work could investigate class-specific tuning, spatially aware validation, and additional model diagnostics.

8. Sources

1. Hawaii-Coral-Reef-ML repository, README, notebooks Part1_EDA.ipynb and Part2_knn.ipynb, and saved Outputs/*.csv files, supplied asHawaii-Coral-Reef-ML-main(1).zip.

2. Prepared modelling data are loaded by the notebooks from the Hawaii_RegimesPredictors GitHub repository, includingPredictors_complete_train.txt, Predictors_complete_val.txt, and Predictors_complete_test.txt.